FH | JOANNEUM
eHealth

Artificial Intelligence in Health Care

Exercise 1

EHT25 WS 2026/27

Readmission Risk from Structured Clinical Data

General Information

Build a first neural network on real clinical data, and compare it honestly to a logistic-regression baseline. You are asked to produce a model whose performance you can describe accurately, including what you cannot conclude from it.

Dataset. *Diabetes 130-US Hospitals for Years 1999–2008*, UCI Machine Learning Repository, dataset 296. 101,766 hospital encounters, 47 features, 130 US hospitals over ten years (`https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008`). Full description of categories for categorical variables can be found in `IDS_mapping.csv`.

Paper: `https://doi.org/10.1155/2014/781670`

```
# load dataset
from ucimlrepo import fetch_ucirepo

diabetes = fetch_ucirepo(id=296)
X = diabetes.data.features
y = diabetes.data.targets
```

1 Problem Formulation [1.5P]:

Define the following **in your own words, before writing any modelling code**:

- The **clinical question**, in one sentence.
- The **target variable**. The raw column `readmitted` has three levels: `<30`, `>30`, `NO`. State your binarisation explicitly and justify it. Different choices produce different published results; yours has to be plausible.
- The **population** used (which encounters you include and exclude, with reasons).
- The **intended user**, and **one decision** the prediction would inform.

2 Data Exploration [2.5P]:

- Structure, distributions, class balance.
- **Encounters vs patients:** count encounters per `patient_nbr`. Report the distribution and state, in one sentence, what it implies for how you must split the data.
- **Missingness:** characterise `weight`, `medical_specialty`, `payer_code` and others. Try to classify the mechanism which leads to missing values and say whether the absence is itself informative (use the paper of the dataset as hint). Decide how to impute missing values, to drop, or to introduce a `IS_MISSING` category.

- **Target:** Check the description of categories of the `discharge_disposition_id`. Account for discharge dispositions that mean the patient cannot be readmitted (e.g. expired, hospice).
- The ICD-9 codes (`diag_1`, `diag_2`, `diag_3`) are high-cardinality. Group them and justify your grouping scheme.

3 Baseline: Logistic Regression [2P]:

Fit a logistic regression on a defensible feature set. Report it honestly as the number the neural network has to beat.

4 Neural Network and Regularization [3P]:

Train a Keras multilayer perceptron on the same data and the same split.

- Vary **capacity** (number of hidden units and hidden layers) and **at least two regularizers** (weight decay, dropout, early stopping) systematically.
- Present **learning curves** for every configuration, and *interpret* them in the text: say what each curve told you and what you changed as a result. Try also different optimizer.
- Select on the **validation** set. Touch the **test** set exactly once, at the end.
- Seed with `keras.utils.set_random_seed(SEED)`. An unseeded comparison is not a comparison. If a change moves your score by less than the run-to-run variation, you have learned nothing.

5 Comparison and Limitations [1P]:

Report **AUROC** and a **confusion matrix** for both models on the **same split**. Does the network beat logistic regression by an amount that means anything? **State explicitly what these numbers do not establish.**

6 Random versus Grouped Split [1P* Bonus]:

Train the same model twice: once with a random split over rows (encounters), once with a split grouped by `patient_nbr`. Report both scores and quantify the difference. Explain the direction of the change and interpret the result.